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Next Generation 7 Tesla Arterial Spin Labeling With Rotated Spiral Acquisition Enables Mesoscale Resolution in 3D Brain Perfusion and Functional MRI

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ABSTRACT

Purpose: To achieve high resolution (≤ 1 mm isotropic) whole-brain perfusion imaging at 7 T with next generation ASL pulse sequence, reconstruction algorithm, and MRI hardware.

Methods: We capitalized on three major innovations: (1) FLASH-based pseudo-Continuous ASL (pCASL) sequence with rotated golden-angle stack-of-spirals (rGA-SoS) sampling; (2) dynamic compressed sensing (CS) reconstruction with high spatiotemporal resolution and motion-resolved self-navigation; and (3) high density array coil and high-performance Impulse gradient of the NexGen 7 T scanner. Whole-brain laminar perfusion imaging was validated by correlation with histological data of microvascular and cell body density, as well as through finger-tapping (FT) and working memory (WM) fMRI tasks.

Results: The proposed rGA-SoS sequence achieved a 3.3-fold SNR and 2-fold higher intraclass correlation coefficient (ICC) compared to matched Cartesian sampling at 7 T, enabling up to 0.8 mm isotropic spatial resolution and/or a temporal resolution of 14 s at 1 mm isotropic. Resting-state perfusion showed strong correlations with microvascular and cell body density. Laminar perfusion fMRI revealed a two-peak activation in the primary motor cortex induced by FT, and distinct laminar profiles for task-positive and task-negative networks during WM task.

Conclusion: This method offers a noninvasive imaging tool to bridge the gap between mesoscopic MRI with microscopic cellular imaging, as well as to investigate neural excitation and inhibition underlying positive and negative fMRI activations.

1 | Introduction

The development of ultrahigh field (UHF) MRI was primarily driven by the desire to image the brain on the mesoscale (e.g., sub-mm) to fill in the gap between microscale on synapses and neurons, and macroscale on brain regions and networks [1, 2]. UHF fMRI with blood oxygen level dependent (BOLD) and vascular space occupancy (VASO) contrasts allowed studying human brain circuit function at the level of cortical layers and columns [3–8]. Despite considerable progresses, challenges remain to link mesoscopic fMRI with underlying neuronal activities: (1) The BOLD signal reflects complex interplays between cerebral blood flow (CBF), cerebral blood volume (CBV), and oxygen metabolism (CMRO2), resulting in limited capability for

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quantitative interpretations of the BOLD signal across cortical layers or depths with different baseline physiology [9, 10]; (2) The BOLD signal is susceptible to venous contaminations such as the pial veins on the cortical surface that confounds laminar fMRI [4, 11–13]; and (3) VASO based CBV fMRI offers improved specificity but only measures relative CBV changes that may be confounded by different baseline CBV values across cortical layers or columns [9].

CBF or microvascular perfusion measured by arterial spin labeling (ASL) is a key parameter for in vivo assessment of neurovascular function. The ASL signal is localized close to the site of neural activation as most of the labeled arterial water exchanges with tissue water in capillaries [14]. Optical imaging in animals provides evidence that CBF can be precisely and rapidly regulated on an extremely fine scale at the level of capillaries (e.g., through pericyte regulation) [15–18]. It has been shown that ASL perfusion fMRI is able to visualize orientation columns in cat visual cortex with superior spatial resolution compared to BOLD fMRI [19]. The hemodynamic response of perfusion signals has also been shown to arise ~1 s earlier than that of BOLD signals [20]. UHF ASL has the dual benefits of increased SNR ($\propto B0^{1.65}$) and prolonged tracer half-life (blood T1 = ~2.1 s at 7 T) [21, 22], resulting in ~5 fold theoretical SNR gain from 3 to 7 T ASL [23, 24]. Since ASL signal is quantitative and originates from capillary/tissue, high resolution 7 T ASL offers a mesoscale perfusion metric that is directly linked to capillary density and neural activity. We and others have demonstrated the feasibility of high-resolution ASL using 3D zoomed GRASE at 7 T [25, 26], turbo-FLASH (TFL) readouts at 7 T [27, 28], and EPI-based acquisitions at 7 T [29, 30], as well as high-resolution laminar ASL at 3 T [31]. However, existing techniques can only image part of the brain, the cerebral cortex with lower resolution (2 mm isotropic), or inefficient labeling methods such as pulsed ASL, limiting their use for fine-grained neuroscience research and clinical diagnosis.

Here we present next generation ASL for whole-brain mesoscale perfusion imaging at 7 T through 3 major innovations: (1) A novel FLASH-based pseudo-Continuous ASL (pCASL) sequence with rotated golden-angle stack-of-spirals (rGA-SoS) sampling; (2) dynamic compressed sensing (CS) reconstruction with high spatiotemporal resolution and motion-resolved self-navigation; and (3) high density array coil and high performance Impulse gradient of the NexGen 7 T scanner. We performed theoretical simulations, test–retest experiments, and systematic comparisons with matched Cartesian-based sequences at 3 and 7 T to demonstrate the significant performance boosts using the proposed next generation ASL technique. By correlating laminar profiles of resting CBF with the BigBrain atlas of cell body density as well as histological data of microvascular density, we show that our technique provides a means to bridge the gap between mesoscopic MRI with microscopic cellular imaging. Finally, we applied laminar perfusion fMRI during finger tapping (FT) and working memory (WM) tasks to demonstrate that our technique is able to differentiate somatosensory input and spinal output in the motor network, as well as to delineate distinct laminar profiles for task-positive and task-negative networks during WM tasks.

2 | Methods

2.1 | Pulse Sequence Design

As shown in Figure 1a, a segmented stack-of-spirals approach was implemented for the FLASH-based rGA-SoS sequence, in which each segment acquires an in-plane (kx-ky) spiral shot and covers all through-plane (kz) samples using centric kz ordering. Label/control acquisitions incorporate randomly sampled kz samples and GA-rotated spiral trajectories (137.5°) with variable density (VD). The full set of spiral segments was divided into multiple subsets, each contributing to different temporal frames for dynamic CS reconstruction. GA rotation was applied both across the through-plane (kz) direction and between segments within each subset, introducing spatial and temporal incoherence in the sampling pattern. Fully sampled central k-space of M0 acquisition was used for coil sensitivity estimation. Figure 1b outlines the pulse sequence within each segment, including pCASL labeling, an optimized background suppression (BS) module, and a FLASH readout. The labeling module (Figure 1c) uses B0-robust parameters (RF duration = 300 us, duty cycle = 500 us, G_{ave} = 0.5 mT/m, Gratio = 12, nominal FA = 15°, balanced scheme) with a Chebyshev RF pulse that provides reduced sidelobes to minimize interference artifacts. Labeling efficiency (LE) was further optimized using a fast prescan with phase-cycling to estimate the global off-resonance across major feeding arteries and adjusting the phase increment accordingly (details in Section S1). The designed BS scheme reduced LE loss due to imperfect inversion (only 1 BS pulse after labeling with the optimized adiabatic inversion pulse) and achieved 5%–20% tissue suppression (see Section S2). Each signal acquisition (Figure 1d) begins with a short TE achieved by combining an asymmetric Shinnar–Le Roux (SLR) RF pulse (designed with maximum phase) and center-out spiral readout.

2.2 | Spiral Imaging Corrections

Several strategies were employed to reduce artifacts associated with spiral sampling (see Figure 2). Gradient impulse response function (GIRF) measurements using 12 trapezoidal gradient blips were used to correct potential distortions in spiral gradient waveforms [32, 33]. The gradient blips used slew rate of 180 mT/m/ms and had durations ranging from 100 to 320 µs in 20 µs steps. A 4-step subtraction method, combining ± gradient polarities and ± slice locations, was used in GIRF measurement to remove systematic baseline shift and B0 eddy current [34]. The GIRF-predicted gradient waveforms can be obtained by a convolution between nominal waveforms and GIRF in time domain. Alternatively, actual spiral gradient waveforms can also be directly measured by dynamic field camera (DFC) (Skope MRT, Zurich, Switzerland) in a separate scanning session (see Section S3). Prior work has shown that both approaches can substantially improve non-Cartesian trajectory fidelity [35]. In this study, GIRF-based correction was applied to data acquired on the Terra system, whereas DFC-based correction was applied to data acquired on the NexGen 7 T system.

Multi-frequency interpolation (MFI) [36] was applied to address blurring from off-resonance by reconstructing the image across nine frequencies from −300 to 300 Hz with 66 Hz steps, then

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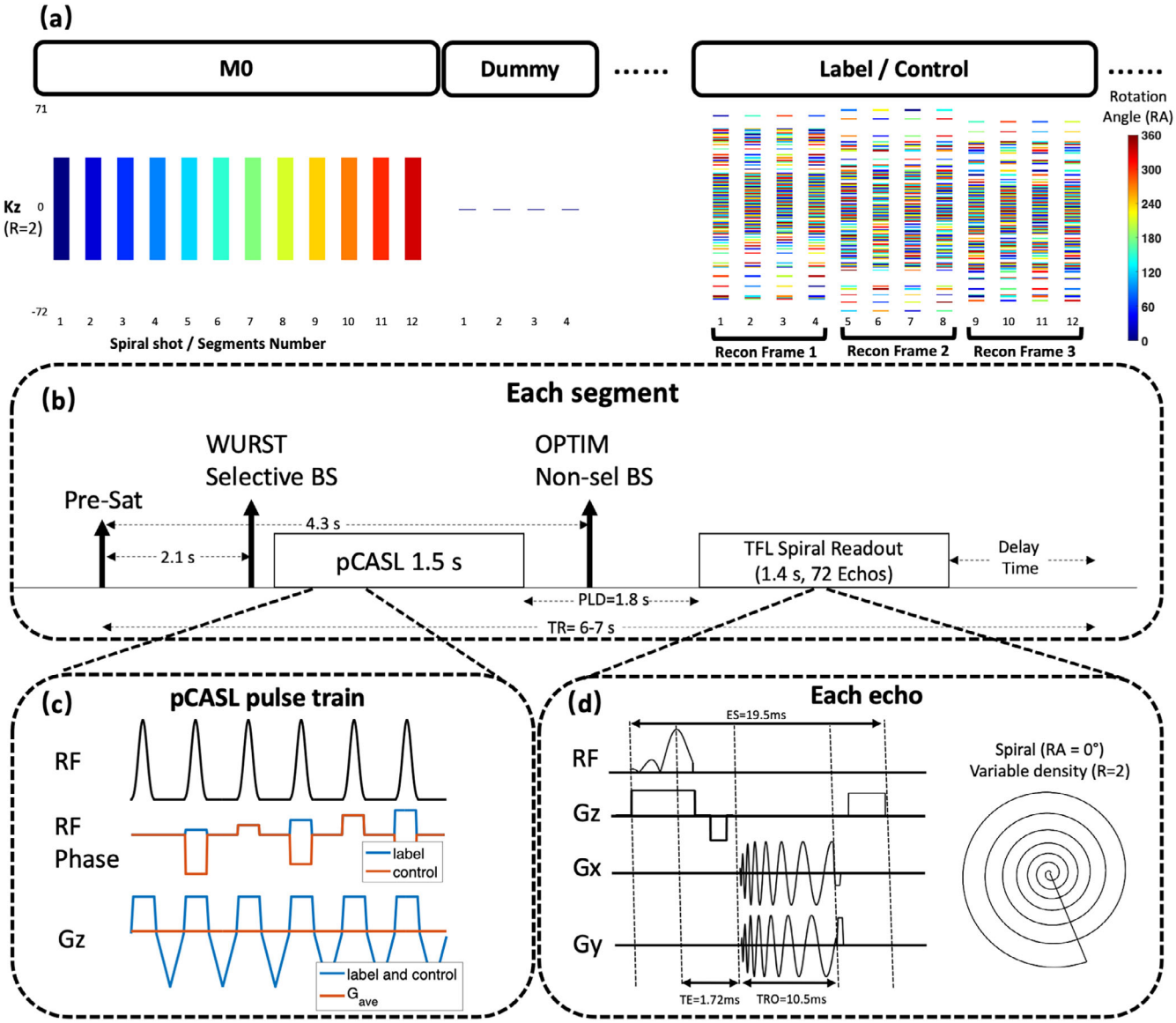


FIGURE 1 | Diagram of proposed rGA-SoS pCASL sequence. (a) The segmented rGA-SoS readout is used for M0, label, and control acquisitions. Each segment samples a spiral shot in-plane (kx-ky) and full kz using centric ordering. Label/control acquisitions apply random kz sampling and spiral trajectories rotated by angles represented by colors. Segments are grouped into dynamic frames for improved temporal resolution and motion robustness. (b) Each TR includes pCASL labeling, background suppression (BS), and a readout segment. (c) The pCASL pulse train uses a B0-robust configuration, and RF phase is adjusted based on a fast prescan. Chebyshev pulses are employed to mitigate sidelobe interference of pCASL labeling. (d) Each echo of the FLASH readout uses short TE via asymmetric SLR pulses and center-out spirals. The SLR pulse also features a 1000 Hz bandwidth to prevent chemical shift artifacts related to fat signals. The gradient waveforms form a spiral trajectory with variable densities and an effective acceleration factor $R = 2$. Spiral readout occupies a larger portion of echo spacing than Cartesian readout, which increases SNR efficiency.

combining them with complex weights estimated from the B0 map (see Section S4).

An asymmetric SLR excitation pulse with a 1000 Hz bandwidth (used for slice selection in the FLASH readout; Figure 1d) was chosen so that fat lies outside the excited slab profile (see Section S5), thereby preventing fat-ringing artifacts during the spiral readout., This pulse design also avoids the prolonged TE, increased scan time, and higher SAR associated with more complex pulse design such as binomial pulse, spectral-spatial pulses, or Dixon-based methods.

2.3 | Dynamic CS Reconstruction

Stack-of-spirals trajectories with VD are inherently more robust to motion than Cartesian trajectories as each segment starts from and oversamples the k-space center region. However, head motion between segments may still degrade image quality with sub-mm resolution. Here we integrated CS reconstruction with motion-resolved self-navigation into the proposed ASL framework, enabling motion-robust submillimeter perfusion imaging [37]. In this framework, each dynamic frame was reconstructed using several consecutive spiral segments (e.g., 4 of the

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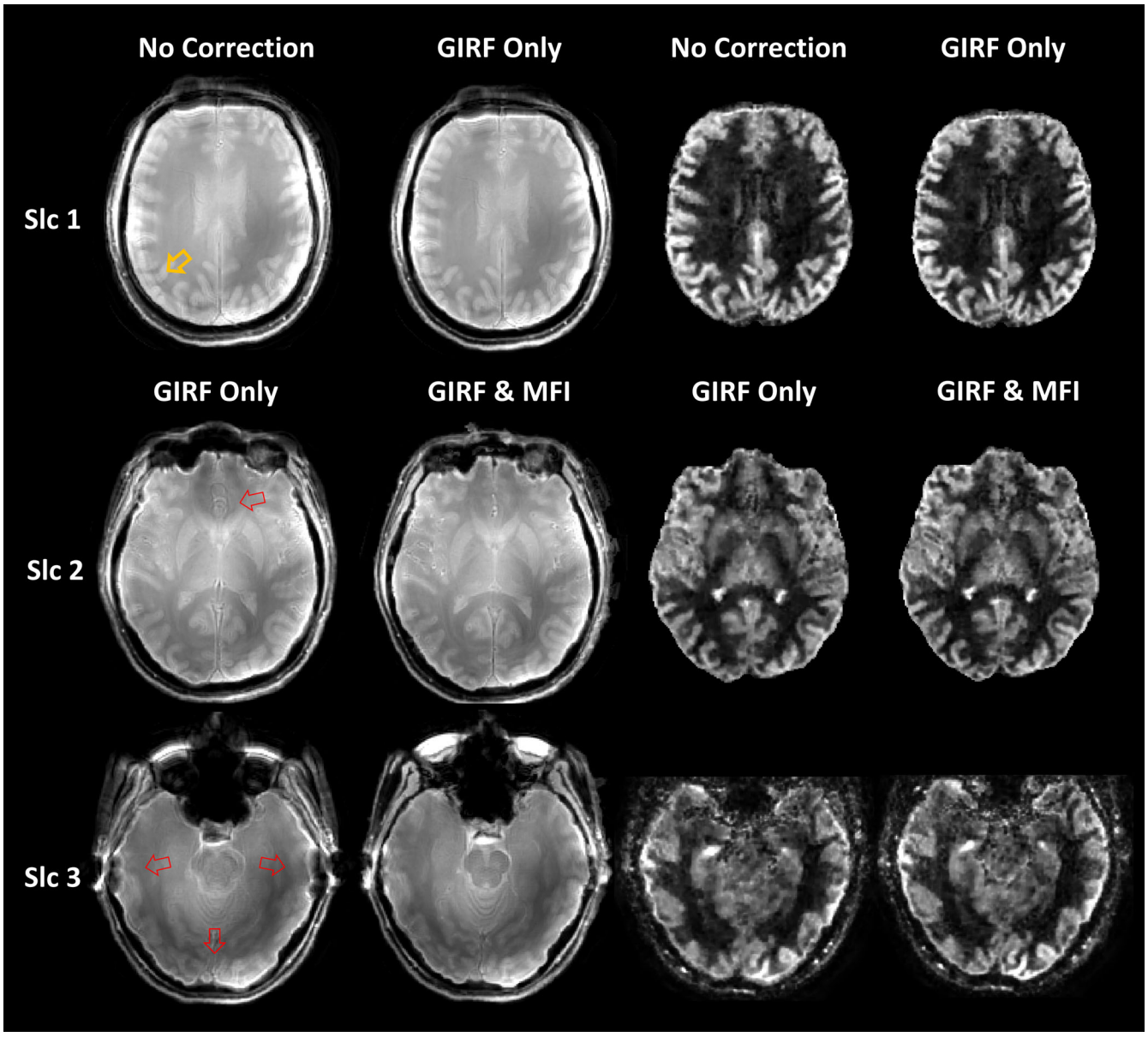


FIGURE 2 | Impact of GIRF and MFI corrections on M0 and perfusion-weighted images (PWI). Three axial slices from a representative subject show the effects of GIRF and MFI corrections. In Slice 1, compared to the image without any corrections, GIRF correction alone substantially reduced ringing artifacts in the M0 image (yellow arrows). In PWIs, the improvement is less visually prominent due to strong intrinsic GM/WM contrast. In regions prone to off-resonance (red arrows), such as the frontal (Slice 2) and temporal lobes (Slice 3), MFI further mitigated blurring artifacts when applied on top of GIRF correction.

12 TRs per label/control acquisition; Figure 1a), enabling retrospective rigid-body motion correction across frames. When combined with the high-performance gradient system, this configuration allowed single-shot label/control acquisitions and an effective temporal resolution of ~14 s for perfusion-based fMRI (see Section 3.6 and Section S13).

Dynamic CS reconstruction incorporates spatial regularization on total variation (TV_r) and temporal regularization on finite difference (FD_t) across dynamic frames to exploit 4D sparsity enabled by the sequence, as formulated in Equation (1),

$$\arg\min_{x(\boldsymbol{r},t)} \| A(x) - \boldsymbol{b} \|_2^2 + \lambda_r \| TV_r(x) \|_1 + \lambda_t \| FD_t(x) \|_1 \quad (1)$$

where x and b represent 4D image and k-space data, respectively. The forward operator A incorporates ESPIRiT-based coil sensitivity weighting [38], non-uniform fast Fourier transform (NUFFT) implemented via fiNUFFT library [39], and k-space undersampling. Spatial and temporal regularization parameters λ_r and λ_t were used to control the trade-off between improved SNR and reduced aliasing artifact versus increased spatial and temporal smoothness. The inverse problem was solved using alternating direction method of multipliers (ADMM) [40], as detailed in Table S1. To ensure consistent scaling across reconstructions, operator A and vector b were normalized to have a unit norm. An augmented Lagrangian penalty parameter $\rho = 0.3$ was used to balance primal and dual residuals. Reconstruction was performed with an in-house Python program using 10 ADMM

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iterations with 3 nested conjugate gradient steps per iteration, requiring ~90 s and 14GB VRAM on a Nvidia RTX 5090 for each label/control image.

Reconstruction parameters were optimized using a digital phantom simulating cortical laminar CBF distributions with realistic noise and rigid-body motion. The model incorporated BS scheme and ASL signal evolution to generate control and label images. Parameter sweeps over flip angles (FA) and regularization weights were performed to identify configurations balancing spatial sharpness, SNR, and motion robustness (see Section S7 for details). The analysis showed that a spatial regularization $\lambda_r = 0.001$ and temporal regularization $\lambda_t = 0.002$ achieved the best trade-off between noise/artifact suppression and spatial/temporal sharpness. These parameters were therefore fixed for in vivo reconstructions.

2.4 | Imaging Protocols

Under the approval from local Institutional Review Boards, a total of 16 subjects (7 males, 27.1 ± 4.4 y/o) participated in MRI experiments conducted on three different Siemens (Siemens Healthineers, Erlangen, Germany) scanners: (1) MAGNETOM Terra 7 T scanner (8Tx/32Rx); (2) MAGNETOM Prisma 3 T scanner (1Tx/32Rx); (3) NexGen 7 T MAGNETOM Terra Impulse edition (16Tx/64Rx). All 7 T scans were performed with circularly polarized (CP) mode on the first level with SAR varying from 75% to 95% across subjects of different genders and body sizes. Six subjects (#1–6) underwent test–retest and 3 T–7 T comparisons, two (#7–8) performed finger-tapping (FT) tasks on Terra, six (#9–14) performed working-memory (WM) tasks on Terra and NexGen 7 T respectively, and two (#15–16) were scanned on NexGen 7 T to demonstrate 1 and 0.8 mm isotropic ASL. Typical scan durations were ~10 min for 1 mm, ~16 min for 0.8 mm static ASL, ~25 min for FT, and ~36 min for WM experiments. A summary of participants and scan times can be found in Table S2.

On Terra 7 T scanner, whole-brain 3D pCASL imaging with 1 mm isotropic resolution was performed using the rGA-SoS sampling scheme with the following parameters: FOV = 256 × 256 × 144 mm^3 and matrix size = 4200 × 12 × 144. An effective in-plane acceleration factor (R) of 2 was achieved via the VD spiral interleaves. Along the through-plane direction, 144 kz partitions were acquired using an echo train length (ETL) of 72, corresponding to a through-plane R of 2. Each M0, label, and control acquisition consisted of 12 spiral interleaves acquired over 12 TRs. Three dynamic image volumes were reconstructed for each label/control acquisition. Additional parameters were: TR = 7 s, effective perfusion weighted image (PWI) volume TR (TR_{PWI}) = 56 s, ES = 19.5 ms, TE = 1.72 ms, ADC bandwidth = 400 kHz, ADC duration = 10.5 ms, and FA = 15°. Note that the number of spiral interleaves was selected to balance ADC duration and T2* blurring, while avoiding excessive acceleration that could degrade image quality. pCASL labeling was performed using the parameters described in Section 2.1 with a post labeling delay (PLD) = 1.8 s and a maximum allowable labeling duration (LD) of 1.5 s by the transmit coil configuration. A matched Cartesian version of the FLASH-based pCASL sequence was implemented using VD elliptic Poisson-disc sampling in the ky–kz plane (see Section S9 for detailed sampling strategy). To achieve the same 1 mm isotropic resolution within comparable scan time, Cartesian sampling required a higher acceleration rate ($R \approx 8.9$) than rGA-SoS due to its lower acquisition efficiency. Imaging parameters included FOV = 224 × 192 × 144 mm^3, matrix = 168 × 192 × 144, asymmetric echo factor = 75%, ES = 5.3 ms, TE = 2.3 ms, ADC bandwidth = 175 kHz, FA = 8°, 12 segments, TR = 7 s, and TR_{PWI} = 168 s.

For additional comparison, traditional GRASE pCASL at 3 T and zoomed GRASE pCASL at 7 T were performed with matched scan times. Imaging parameters of 3D GRASE pCASL were: resolution = 2.5 × 2.5 × 3 mm^3, FOV = 240 × 240 × 132 mm^3, matrix = 96 × 64 × 44, TR/TE = 4.1 s/36.5 ms, EPI factor = 63, turbo factor = 12, 4 segments, bandwidth = 250 kHz, FA = 120°, and 152 TRs including one M0, four dummy TRs, and 18 label/control pairs. Labeling parameters tailored for 3 T were [41]: LD = 1.8 s, PLD = 1.8 s, RF duration = 500 us, duty cycle = 920 us, G_{ave} = 0.6 mT/m, Gratio = 10, FA = 25°, balanced scheme. For zoomed GRASE pCASL at 7 T, 1 mm isotropic resolution was achieved with the following parameters: FOV = 100 × 50 × 24 mm^3, matrix = 100 × 50 × 18 with 6/8 partial Fourier, TR = 2.8 s, TE = 26.8 ms, EPI factor = 25, turbo factor = 6, bandwidth = 150 kHz, FA = 120° with initial stabilization, ES = 0.79 ms, 6 segments per label/control, and TR_{PWI} = 33.6 s. These reference scans served as benchmarks for evaluating image quality and readout performance across field strengths. All ASL protocols were summarized in Table S3 (Section S10).

The anatomical data was acquired using MP2RAGE with 0.7 mm isotropic resolution, FOV = 224 × 224 mm^2, and 224 sagittal slices. Sequence parameters were: slice partial Fourier = 6/8, FA = 4°/5° (for inversion times TI1/TI2), TE/TI1/TI2/TR = 3.43/1000/3200/4500 ms, GRAPPA acceleration factor = 3 with 37 phase-encoding reference lines. The total scan time was 9 min and 51 s.

2.5 | Next Generation 7 T System and ASL Protocols

The NexGen 7 T system offers an unprecedented brain imaging platform by integrating high-density (up to 128-channel) RF arrays, high-performance Impulse gradients and state-of-the-art pulse sequences at 7 T [42]. In particular, the Impulse gradient coil offers 2.5 times magnitude (200 mT/m) and 4.5 times slew rate (900 T/m/s) compared to standard 7 T Terra (80 mT/m, 200 T/m/s) while minimizing peripheral nerve stimulation (PNS), thereby offering the opportunity to dramatically improve the spatiotemporal resolution of 7 T ASL. In addition, DFC equipped on the system ensures a robust reconstruction under high-performance gradient switching.

On NexGen 7 T scanner, three different protocols of rGA-SoS pCASL, demonstrating advantages from different perspectives, were performed using higher gradient limits than Terra (Gmax = 72 vs. 20 mT/m, SR = 480 vs. 190 mT/m/ms): (1) *Protocol 1. Improved temporal resolution* — Matched the Terra protocol in resolution and FOV but requires only four spiral interleaves for each label/control/M0 acquisition, resulting in TR_{PWI} = 14 s;

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(2) *Protocol 2. Improved SNR*—Based on protocol 1, but with an added kz segment and increased FA from 15° to 23°, as demonstrated by Figure 7b. (3) *Protocol 3. Improved spatial resolution*—Had a FOV of $256 \times 256 \times 140\,mm^3$ and 8 spiral interleaves, divided into 4 dynamic frames. Each label/control acquisition covered 176 kz partitions using 2 kz segments (same as protocol 2). Note Gmax was first determined by the FOV and readout dwell time, but in practice we used a more conservative Gmax to reduce trajectory-related artifacts. Although the Impulsegradient coil supports up to 900 mT/m/ms slew rate, the operational slew rate was set lower to avoid excessive gradient heating during long spiral readouts. DFC was used for all scans performed on the NexGen 7 T scanner.

2.6 | Data Analysis

Preprocessing: MRI data were analyzed using AFNI [43], FreeSurfer [44], and the mripy package [45]. The preprocessing of volumetric ASL data included linear motion correction in control and label images, CBF quantification using a standard kinetic model [46] (see Section S11), and upsampling to a finer grid of 0.5 mm isotropic resolution. T1w anatomical image was registered to CBF volumes. A surface-based approach was used for cortical data analysis. The T1w MP2RAGE anatomical volume was segmented into white matter, gray matter, and cerebrospinal fluid.

Correlating CBF with CSI: Cell body staining intensity (CSI) data were obtained from open-access BigBrain [47] repository (https://bigbrain.loris.ca/main.php). The donor brain was sectioned into 7400, 20 µm thick slices followed by silver-stained to highlight cell bodies and digitized. These images underwent manual inspection, correction and nonlinear alignment to a postmortem MRI of the same individual. Finally, the 3D reconstruction was achieved through a coarse-to-fine hierarchical procedure. To extract laminar profile across whole brain, an equi-volume algorithm was used to calculate the relative depth value of each gray matter voxel via mripy package [45]. Six equivolumetric surfaces between the outer and inner cortical surfaces were generated. The preprocessed CSI and CBF volumetric data were mapped to the normalized inflated cortical surface (std141). Laminar profiles of CSI and CBF in 6 ROIs, including primary visual cortex (V1), primary motor cortex (M1), supplementary motor area (SMA), anterior insula (aIns), dorsolateral prefrontal (dlPFC), and medial superior frontal (MSF) were calculated. The 6 ROIs were selected from different microstructural and functional gradients. To investigate the relationship between layer specific CBF and CSI, Pearson correlation coefficients (R) of the laminar profiles of CBF and CSI across different ROIs were calculated.

fMRI data processing: CBF changes (ΔCBF) maps between task and rest states were obtained through subtraction. For extracting two-peak laminar profile in M1, boundaries of CSF/gray matter and white/gray matter in M1 were manually drawn on control images, and 20 cortical layers were segmented based on equi-volume layering approach using LayNii [48], from which laminar task-induced ΔCBF values were extracted. For the WM task, considering the low temporal resolution of ASL sequences, the design matrix in the general linear model (GLM) analysis directly used a boxcar waveform, not convolved with a sample hemodynamic response function (HRF). Motion parameters, their derivatives and square of derivatives, were included as regressors of no interest. The activation maps presented hide points with voxel-*p* values greater than 0.01, retain areas showing significant positive and negative activations without correction, and filter out clusters with size less than 500. ROIs were drawn on surfaces in five activated brain regions, including dorsomedial prefrontal cortex (dmPFC), middle frontal gyrus (MFG), frontal eye field (FEF), intraparietal sulcus (IPS), and posterior cingulate cortex (PCC). The preprocessed CBF volumetric data were mapped to the inflated normalized cortical surface, followed by surface-based smoothing with a 3 mm Gaussian kernel.

Image quality metrics: Test–retest reliability was assessed using intraclass correlation coefficients (ICC). Voxel-wise ICC of two visits was calculated using 'A-k' type [49]. To calculate SNR, each repetition of PWI was subtracted from the mean PWI to generate a noise map. The signal was defined as the mean PWI intensity within the gray matter mask (threshold = 0.9), and noise was estimated as the standard deviation of the noise map within the white matter mask (threshold = 0.99).

3 | Results

3.1 | FLASH-Based pCASL Sequence With Rotated Golden-Angle Stack-of-Spirals (rGA-SoS) and Variable Density

Compared to standard Cartesian sampling, this spiral-based approach increases FA by a factor of 2 (15° vs. 8°) and increases SNR efficiency by 50% (TRO/ES = 10.5/19.5 ms vs. 1.5/5.3 ms). Additionally, the longer readout duration reduces the total number of TRs required, thereby minimizing perfusion signal decay. These benefits were validated through digital phantom simulations described in Section 2.3 and Section S3. Figure 3a shows the simulations comparing Cartesian and rGA-SoS sampling schemes under matched blurring conditions, and demonstrated that a 15° FA in rGA-SoS achieved equivalent point-spread-function (PSF) sharpness (FWHM = 2 voxels) to an 8° FA in Cartesian sampling, yielding a 3–3.8-fold SNR advantage, as indicated by the numbers in Figure 3b. Motion modeling shown in Figure 3c further confirmed reduced artifact accumulation due to shorter acquisition windows in rGA-SoS.

Based on test–retest studies on 6 healthy volunteers ~1 week apart, rGA-SoS pCASL provided 3.3 ± 1.0 times gain in SNR and ~2 times ICC across all brain regions compared to matched Cartesian based pCASL sequence (Figure 4).

We also estimated the LE at 7 T by comparison with 3 T pCASL, using the relative CBF values across all brain regions under the assumption of LE = 1. Estimated LE at 7 T was 88.75% of that at 3 T, which closely aligns with theoretical estimates of LE = 0.75 and 0.85 at 7 T and 3 T, respectively (Figure 5). With LE = 0.75, CBF at 7 T was measured as 46.6 ± 4.6 and 22.2 ± 2.4 mL/100 g/min in GM and WM, respectively. Additionally, interhemispheric asymmetry in relative LE was found to be less than 8% in all regions. A stronger correlation between CBF and gray matter partial volume ratio (GMpv) was observed at

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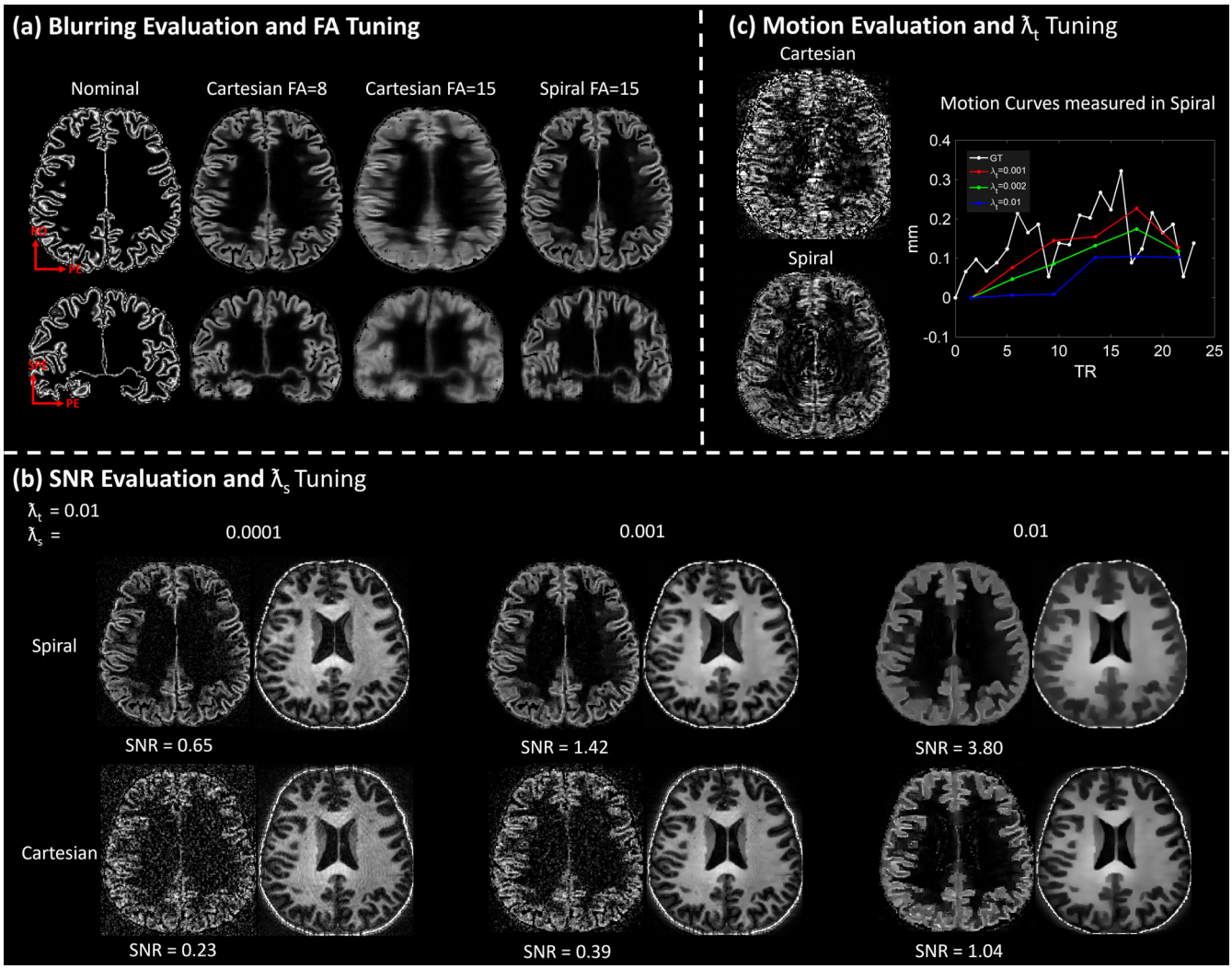


FIGURE 3 | Digital phantom simulations for comparison between rGA-SoS and Cartesian. (a) Simulation of Blurring effects. Spiral FA = 15° and Cartesian FA = 8° achieved comparable sharpness (FWHM = 2), while Cartesian FA = 15° showed excessive blurring (FWHM = 4). (b) SNR improved with increasing λ_r, with Spiral sampling yielding 3–3.8 times higher SNR than Cartesian across all regularization strengths. (c) Motion artifacts were substantially reduced in rGA-SoS due to shorter acquisition windows and the nature of spiral sampling. Motion curves show increased temporal smoothness with larger temporal regularization λ_t, with excessively large λ_t flattening the trajectory by suppressing inter-segment variations.

7 T ($r = 0.98$) compared to 3 T ($r = 0.52$), validating the improved spatial resolution. The high sampling efficiency and LE of rGA-SoS pCASL sequence enabled high quality iso-1 mm resolution whole brain perfusion imaging on a standard 7 T Terra scanner.

3.2 | Dynamic CS Reconstruction With Motion-Resolved Self-Navigation

The proposed dynamic CS reconstruction markedly improved motion robustness compared to Cartesian sampling. Simulation results (Figure 3) and in vivo data (Figure 6) demonstrate that motion artifacts arising from between-segment motion were substantially reduced when temporal regularization ($\lambda_t = 0.002$) was applied to resolve frame-wise motion before motion correction. This approach preserved cortical detail and SNR while effectively suppressing motion artifacts. As shown in Figure 6, moderate temporal regularization captured inter-segment motion that was subsequently removed by rigid-body alignment, whereas overly strong regularization ($\lambda_t = 0.01$) oversmoothed dynamic variations. These results confirm that the motion-resolved CS framework enables artifact-resilient, high-resolution perfusion imaging suitable for high resolution ASL and perfusion-based fMRI.

3.3 | Improving Spatiotemporal Resolution Using NexGen 7 T

As shown in Figure 7, faster and higher magnitude spiral gradient waveforms can be implemented on NexGen 7 T, enabling only one third of spiral shots (4) compared to standard 7 T Terra with 12 shots for iso-1 mm resolution. As shown in Figure 7b, the addition of kz segmentation and increased FA (FA = 23° vs. 15°) led to a 41% increase in SNR without degrading PSF, allowing the resolution to be further improved to 0.8 mm isotropic (Figure 7c and Figure S7). Faster gradient waveforms also improved temporal resolution for perfusion-based fMRI (Section 3.6) and minimized

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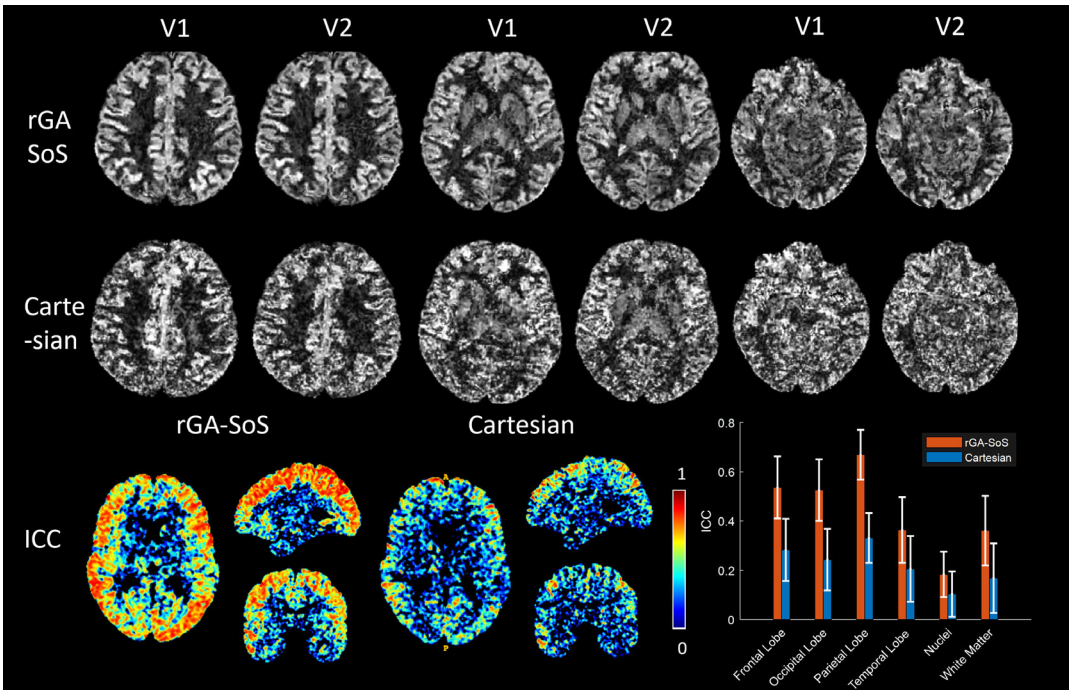


FIGURE 4 | Comparison of test–retest perfusion images between rGA-SoS and Cartesian sampling. Three axial slices from two visits (V1 and V2) are shown for a representative subject. rGA-SoS sampling achieved 3.3 ± 1.0 times SNR, fewer artifacts, and greater anatomical detail than Cartesian sampling, with consistent image quality and perfusion quantification across visits. ICC maps from six subjects further illustrate improved test–retest reliability with rGA-SoS, especially in cortical regions. The bar graph summarizes regional ICCs, showing approximately 2-fold values for rGA-SoS across all regions. ICCs were highest in frontal, occipital, and parietal lobes (˜0.6), while lower values were observed in temporal lobes, subcortical nuclei, and white matter potentially due to lower B1 and/or perfusion.

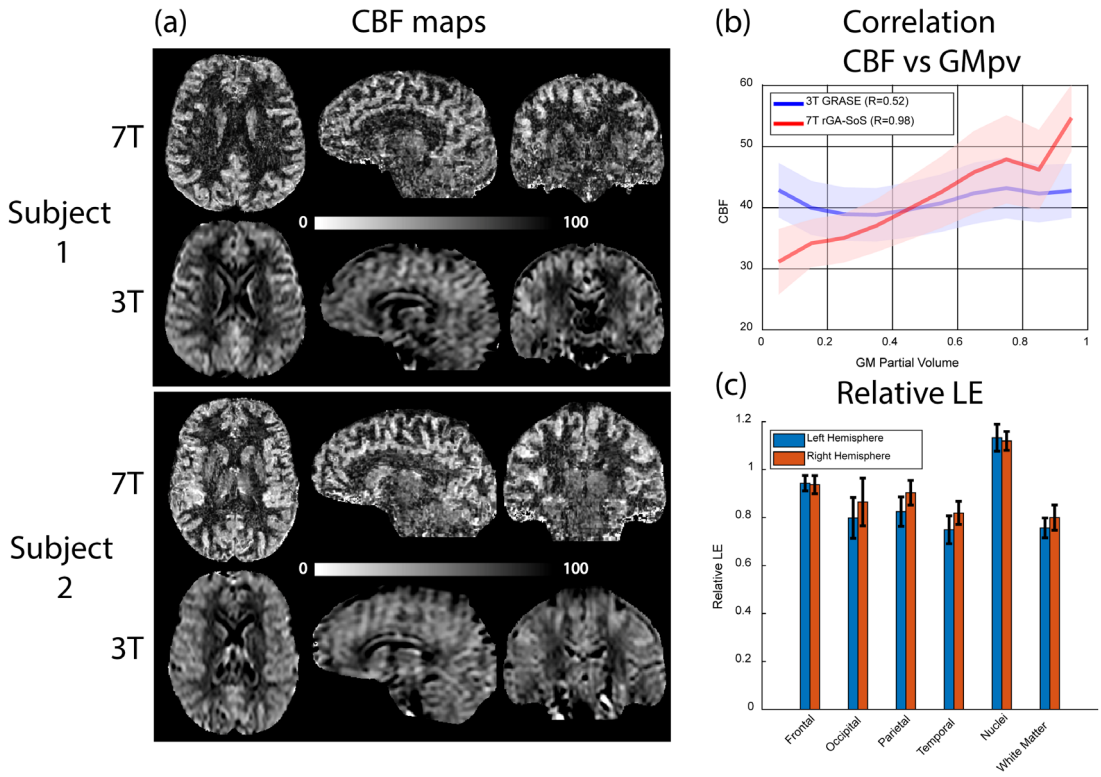


FIGURE 5 | Comparison of 7 T rGA-SoS and 3 T GRASE pCASL in image quality, partial volume (PV) effects, and labeling efficiency (LE). (a) Representative CBF maps from two subjects show improved anatomical detail and reduced PV effects at 7 T compared to 3 T. (b) Correlation between CBF and GMpv significantly improved from $R = 0.52$ at 3 T to $R = 0.98$ at 7 T, indicating reduced PV effect. Shaded areas indicate SD across all voxels within each GMpv bin and across all six subjects. (c) Relative LE values across brain regions show that 7 T achieves comparable LE to 3 T (mean 88.75%), with minor (< 8%) interhemispheric asymmetry.

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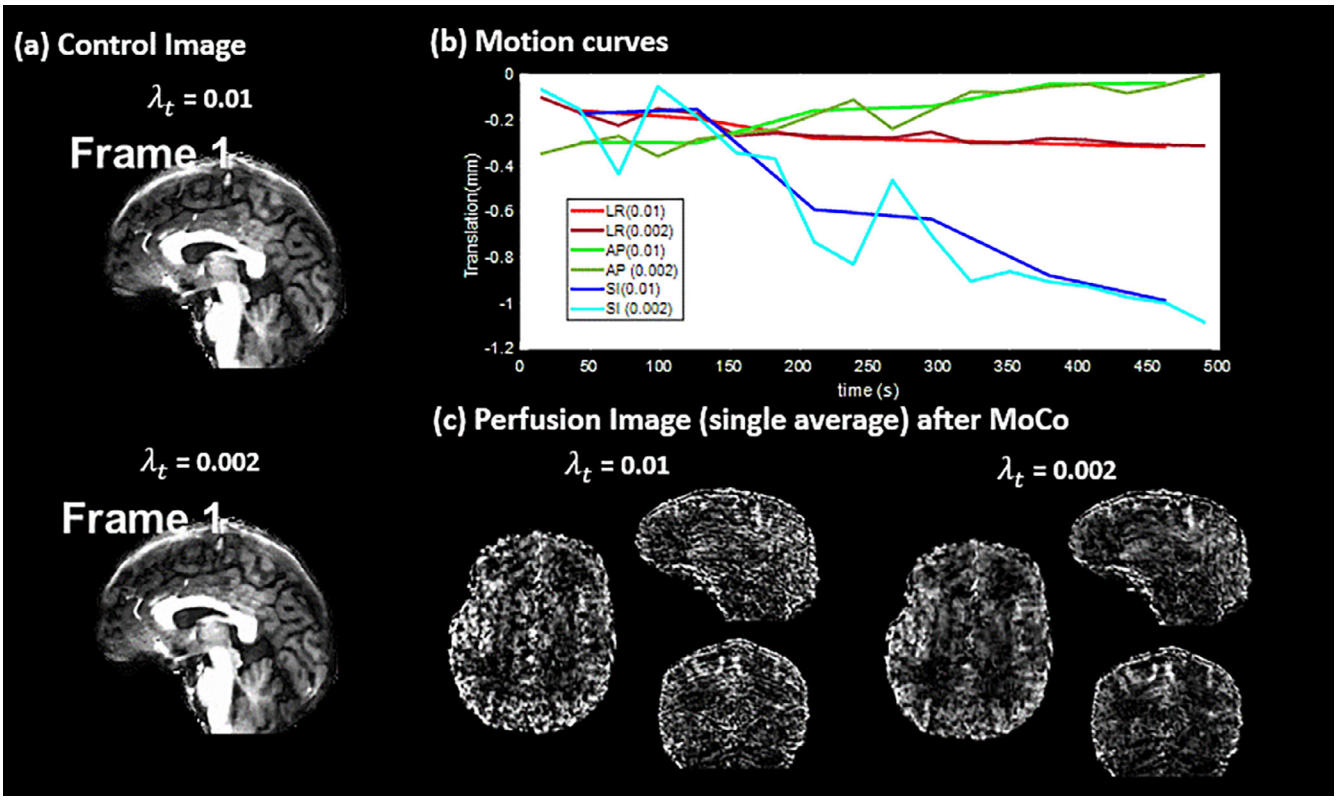


FIGURE 6 | (a) Control images reconstructed by two temporal regularization strengths. Between-segment motion can be captured by an appropriate selection of $\lambda_t = 0.002$ and then minimized by motion correction (see animated GIF file in Supporting Information for Review). (b) Motion curves showing patient motion. (c) A single average PWI at the time of the worst motion occurred showing reduced motion artifacts when between-segment motion can be resolved.

motion artifacts when combined with dynamic CS reconstruction and self-navigation, as each image can be acquired with just 1–2 TRs.

3.4 | Correlation of Laminar Perfusion With BigBrain Atlas of Cell Body Density

We first performed laminar perfusion analysis across the whole brain (Figure 8a), and found that the middle layer exhibited the highest CBF, followed by the superficial and deep layers. However, regional CBF variations can be observed across the three layers, which demonstrate that the proposed technique is capable of capturing regional laminar CBF profiles. For instance, histological study on human brain specimens showed microvascular density is higher in the superficial than deep layers in M1, whereas the opposite pattern was reported in V1 [50]. This is consistent with our measured CBF laminar profiles which peak towards middle to superficial layers of M1, and towards deep layer of V1 (Figure 8b).

Comparison was conducted between laminar profiles of CSI and CBF across cortical layers in six representative ROIs, including V1, M1, SMA, aIns, dlPFC, and MSF. High Pearson correlations between laminar profiles of CSI and CBF were observed in all six ROIs ($r \geq 0.82$, BHp (BH-adjusted FDR-corrected p value) < 0.0015, Figure 8c). Since neuronal cell bodies are metabolically active and require continuous oxygen and glucose delivery, a higher cell body density in a given cortical layer likely reflects increased metabolic demand and thus higher CBF. These findings suggest that laminar CBF may serve as an in vivo surrogate marker for cell body density.

3.5 | Laminar CBF Profile in M1 During FT Task

Figure 9 shows CBF maps acquired with rGA-SoS pCASL in two representative subjects during both resting state and FT task. Corresponding images from Zoomed (inner volume) GRASE pCASL are also included for comparison [25]. Both methods are insensitive to venous signal changes in cortical veins which is a confound present in GE-BOLD, while the proposed method has ˜80 times larger 3D volume size than Zoomed GRASE, potentially offering improved SNR and sensitivity. Robust FT-induced CBF changes (ΔCBF) are evident in M1, as highlighted in the zoomed-in regions. Quantitative ΔCBF values are shown in the accompanying plots. The ΔCBF profiles during FT exhibit a characteristic two-peak pattern in the deep and superficial layers, corresponding to the spinal output and somatosensory input of the motor circuit [6, 25]. Laminar profiles of rGA-SoS (orange zoomed-ins and curves) and Zoomed GRASE (blue) pCASL showed peak ΔCBF values of 41.57 ± 4.90 and 40.63 ± 11.12 mL/100 g/min in the superficial layer, and 37.55 ± 3.40 and 23.90 ± 10.32 mL/100 g/min in the deep layer, respectively. This finding again validated the

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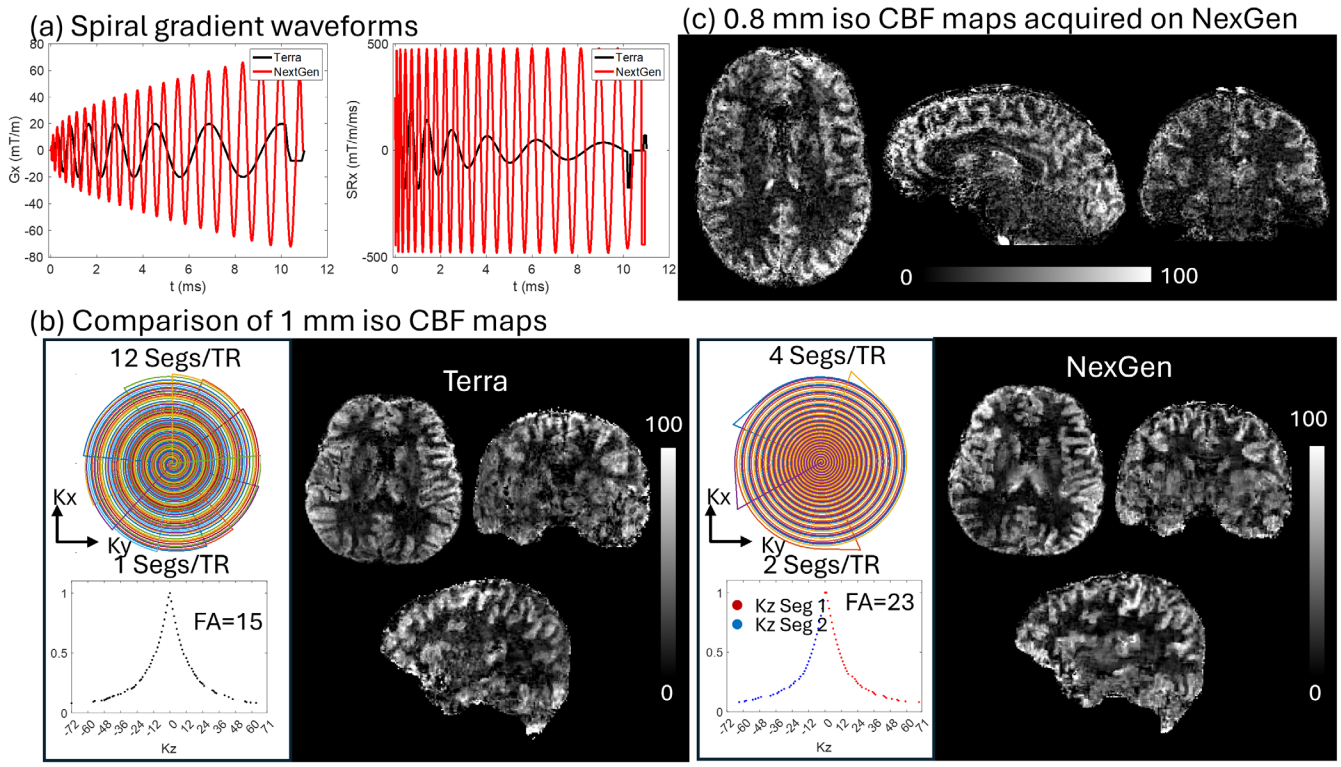


FIGURE 7 | Enhanced spatiotemporal resolution enabled by NexGen 7 T gradients. (a) Spiral gradient waveforms show substantially faster slew rates and gradient amplitudes on the NexGen 7 T scanner, enabling only 1/3 spiral interleaves needed. (b) Comparison of 1 mm isotropic CBF maps. NexGen reduced spiral interleaves from 12 to 4 while maintaining image quality, allowing the addition of kz segmentation and increased FA (FA = 23° vs. 15°) to boost SNR by 41% without compromising slice sharpness. (c) Whole-brain 0.8 mm isotropic CBF maps demonstrate fine structural detail and high CBF values aligned with cortical gray matter (Section S12), illustrating the potential of the proposed sequence and gradient system for submillimeter perfusion imaging.

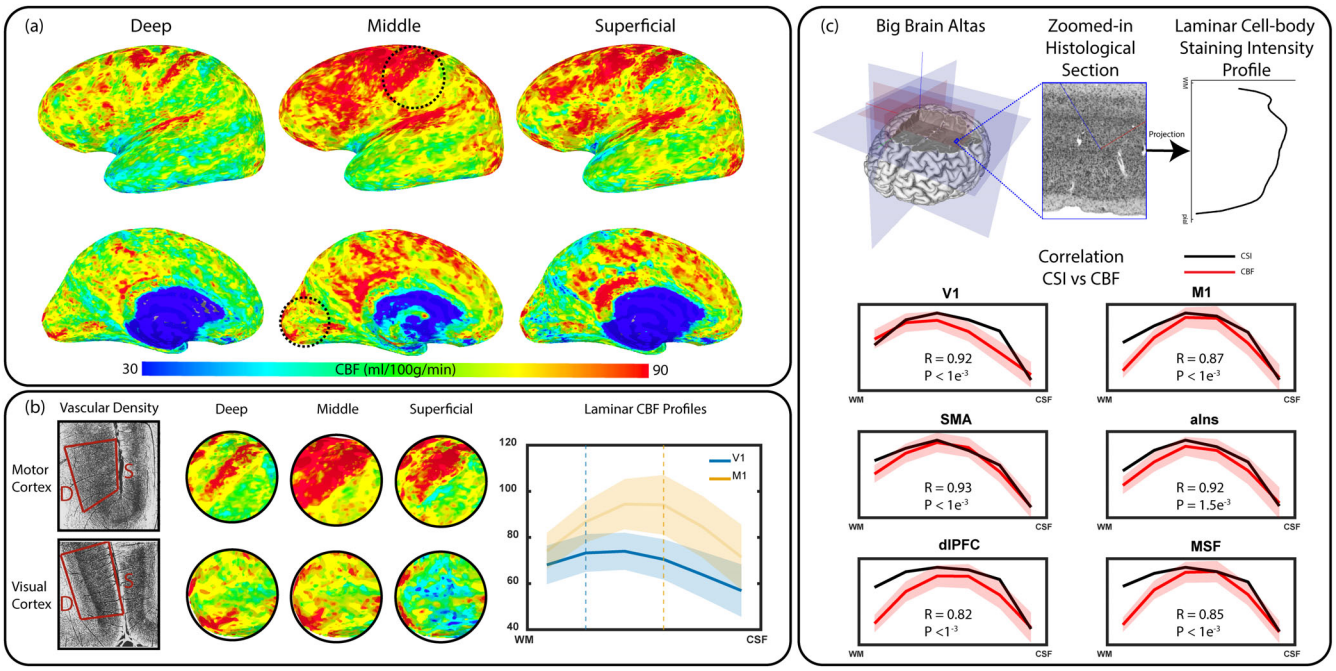


FIGURE 8 | Regional and laminar analysis of CBF distribution on cortical surface and its association with microvascular and cell body density. (a) Cortical surface maps show distinct CBF distributions across deep, middle, and superficial layers, averaged over six subjects. The middle layer exhibited the highest CBF, with region-specific differences observed at each depth. (b) Laminar CBF profiles in the M1 and V1 correspond to known vascular density distributions: M1 shows higher superficial CBF, while V1 shows deeper CBF dominance. (c) Laminar CBF profiles closely match cell staining intensity (CSI) profiles derived from the BigBrain atlas across multiple regions. Strong correlations ($R = 0.82–0.93$, BHp < 0.0015) support the use of CBF as a non-invasive surrogate for laminar neuronal cell body density. Shaded areas represent SD across subjects.

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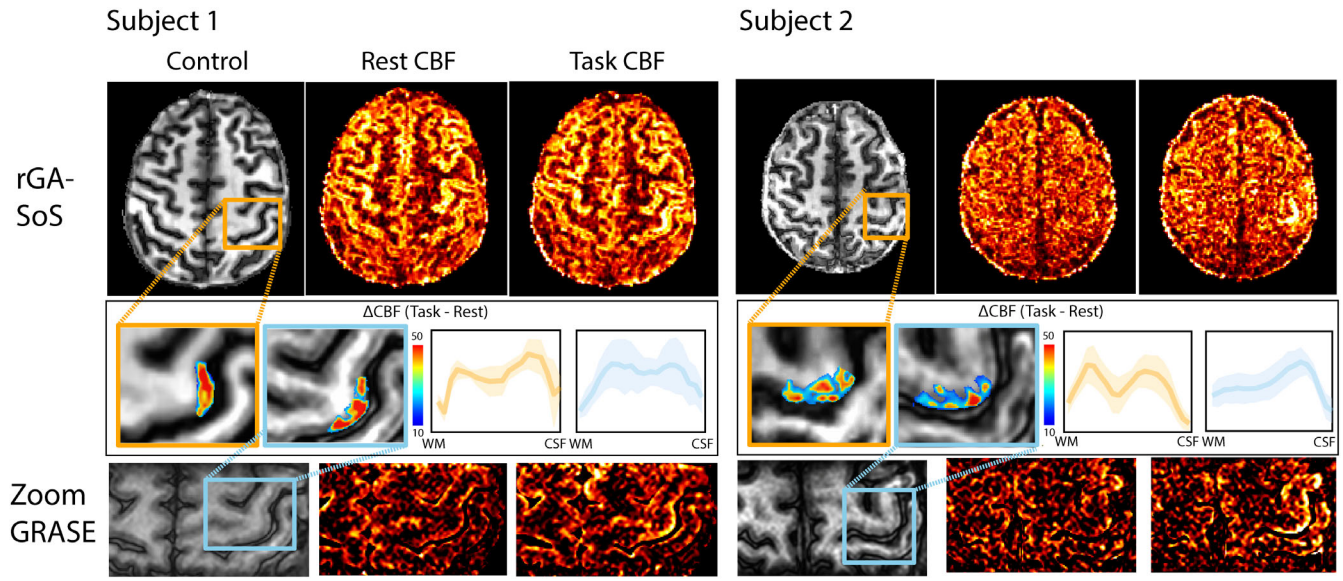


FIGURE 9 | Functional perfusion mapping during finger-tapping (FT) task using rGA-SoS and Zoomed GRASE. Axial images from two subjects show control image, resting-state, and FT task CBF maps using rGA-SoS, with corresponding zoomed GRASE images for comparison. Robust FT-induced ΔCBF responses are observed in the M1, highlighted in zoomed-in regions. Line plots show ΔCBF profiles across cortical depth, revealing a consistent two-peak pattern in both methods, with peaks in deep and superficial layers. Shaded areas represent SD of ΔCBF values at each depth. The rGA-SoS sequence provided improved laminar ΔCBF profiles and wider coverage compared to zoomed GRASE.

functional sensitivity and mesoscale resolution of the proposed technique.

3.6 | Laminar CBF Changes in Task-Positive and Negative Networks During WM Task

Figure 10 shows perfusion fMRI results of 2-back WM tasks (see Section S6 for detailed experiment paradigm) using rGA-SoS pCASL with dynamic CS reconstruction on the standard 7 T Terra and NexGen 7 T, respectively. Characteristic task-positive activations in the frontoparietal network, including dmPFC, MFG, FEF, IPS, as well as task-negative activations in the default mode network (DMN) including PCC can be observed, in line with previous findings [51]. Figure 10b illustrates the laminar relative CBF changes (ΔCBF/CBF) within these regions during the WM task across five subjects scanned on Terra and one subject scanned on NexGen 7 T. A distinct laminar pattern was revealed that deep layer has significantly higher ΔCBF change than superficial layer for both positively and negatively activated brain regions (BHp < 0.05 in IPS, FEF, and PCC). The results acquired on NexGen showed a consistent trend. Figure S9 (Section S13) shows ΔCBF time courses of WM tasks on both scanners, demonstrating four times acceleration on NexGen 7 T with an effective temporal resolution of 14 s per perfusion image compared to 56 s on standard 7 T Terra.

4 | Discussion

In this study, we present a significant advance in mesoscale perfusion imaging through the development of next-generation ASL at 7 T. Beyond prior implementation of spiral ASL at 3 T [52–58], a central novelty of this work lies in the substantial sequence engineering required to make high-resolution ASL feasible at UHF. We adopted a FLASH-based pCASL readout to overcome the SAR limits, T2/T2* shortening, and distortion that challenge GRASE, TSE, and EPI at 7 T. To improve SNR efficiency and motion robustness, we replaced Cartesian sampling with a variable-density stack-of-spirals design and incorporated rotated golden-angle spirals to enhance incoherence for CS reconstruction. Achieving sub-millimeter whole-brain coverage also required a segmented strategy supported by dynamic CS and motion-resolved self-navigation and a high performance gradient system on NexGen 7 T. Additional but essential optimizations, including labeling and BS optimization, trajectory correction, MFI deblurring, and a narrow band asymmetric SLR pulse for fat suppression, were integrated to enable robust performance at 7 T. This work represents the first fully integrated spiral FLASH pCASL framework at 7 T and provides practical insights for future development. These technical advancements bridge the gap between mesoscopic MRI and microscopic cellular imaging, providing both high sensitivity and specificity for functional neuroimaging. The biological relevance of this approach was supported by strong correlations between laminar CBF profiles and cell body density, suggesting its potential as an in vivo biomarker of cortical microstructure. Additionally, the technique resolved layer-specific activation patterns, differentiating sensory input and motor output in a FT task, and revealing distinct laminar profiles in both task-positive and task-negative networks during WM engagement. Together, this study established a framework for mesoscale perfusion imaging, offering a promising approach for fine-grained neuroscience research and clinical diagnosis.

Given our results show quantitative imaging with high spatial specificity, ASL stands out as a promising technique that can particularly benefit from UHF with the potential to gain broader adoption compared to other modalities. However, after

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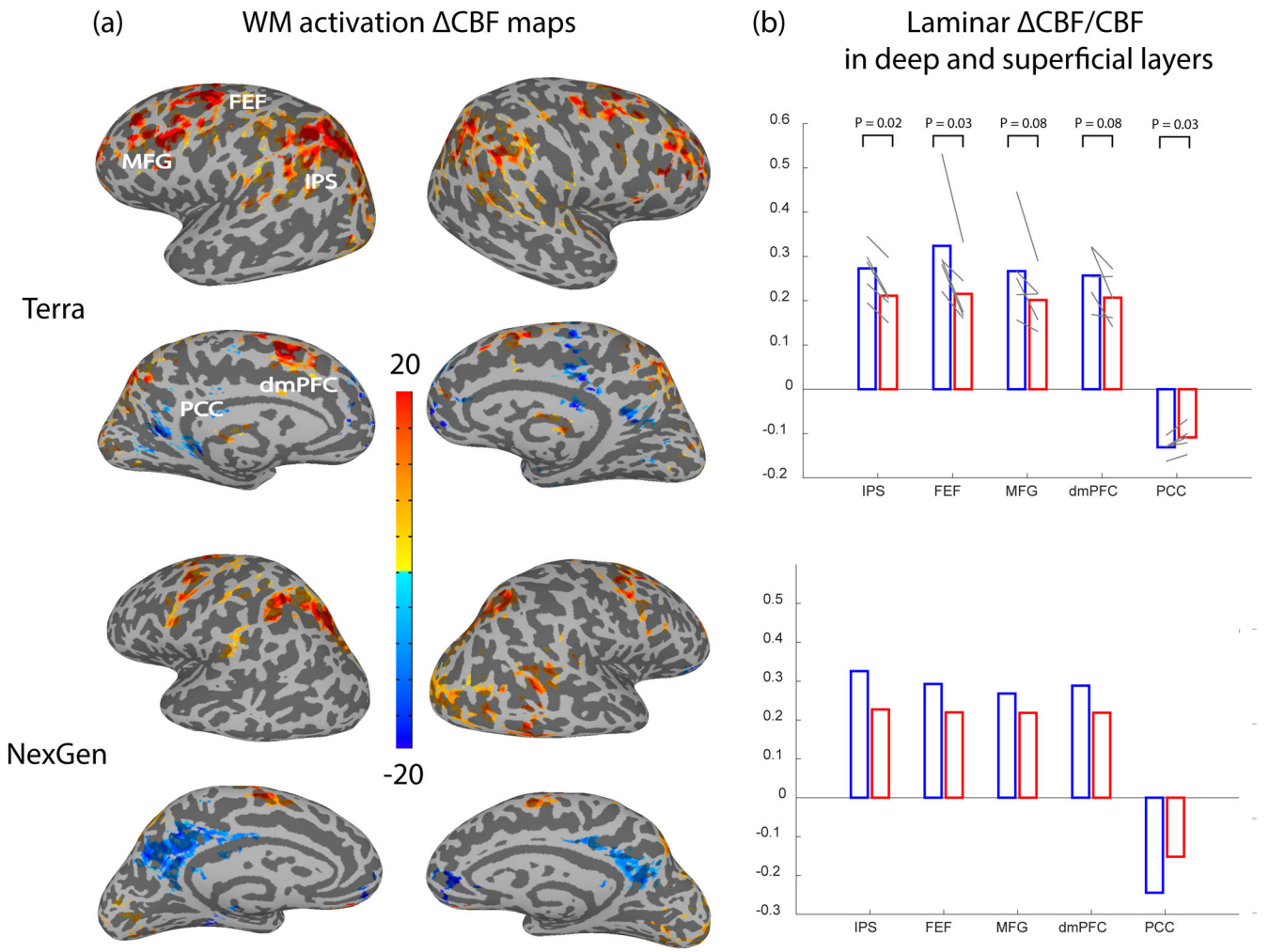


FIGURE 10 | Whole-brain laminar ΔCBF activation during WM tasks. (a) ΔCBF activation maps during a visual 2-back WM task show increased activity in IPS, FEF, MFG, and dmPFC, and reduced activity in PCC, potentially due to task-induced suppression of the DMN. (b) Relative ΔCBF change (ΔCBF/CBF) in selected regions revealed a distinct laminar pattern that deep layer has significantly higher ΔCBF change than superficial layer for both positively and negatively activated brain regions. Gray connecting lines show the within-subject paired values.

a decade, the advent of the next generation ASL has been hampered by several technical challenges. The success of this study builds on a range of innovations and optimizations of the labeling module and readout sequence at UHF. The high, reliable, and spatially homogeneous blood labeling achieved in this study, comparable to that of conventional 3 T ASL, laid the foundation for reducing voxel size at 7 T to 5% (20 times smaller) of that at 3 T ASL. Building upon our prior study [27], this significant voxel size reduction can be achieved because SNR gain does not arise from field strength (˜3-fold) alone, but from multiplicative improvements in acquisition efficiency (˜3-fold) and dynamic CS reconstruction (˜2-fold). Among the three major innovations, the FLASH-based rGA-SoS readout contributed the most to the SNR gain, effectively mitigating SAR limits and distortion associated with EPI/GRASE and yielding a 3.3-fold SNR increase relative to Cartesian sampling. Dynamic CS reconstruction and the high-performance gradients on the NexGen 7 T provided additional improvements in SNR, temporal resolution, and motion robustness. With markedly reduced voxel size and improved acquisition efficiency, ASL was transformed from a modality limited to regional and group-level analysis to a powerful approach for capturing mesoscale perfusion patterns at the individual-level. Note that our 3 T comparison used the standard 1.8 s LD suggested by the ASL White Paper and the widely adopted 3D-GRASE readout [59], which remains recommended at 3 T. We did not implement the rGA-SoS FLASH readout at 3 T in the comparison study because it was specifically designed for 7 T, where EPI/GRASE is limited by distortion and SAR. These constraints do not apply to 3 T, where GRASE remains the appropriate high-performance reference sequence. However, the sampling and reconstruction strategies are directly applicable to 3 T and could be combined with TSE sequence to improve the quality of ASL.

Dynamic CS reconstruction with motion-resolved self-navigation represents a major advance in achieving motion-robust, high-resolution perfusion imaging. By subdividing the acquisition into short temporal segments and reconstructing individual dynamic frames with spatiotemporal regularization, this approach enables retrospective correction of head motion from imaging data. Such motion resilience is critical for achieving sub-mm resolution, capturing subtle and localized perfusion details, especially in

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studies involving extended scan time, multiple runs, or populations prone to movement. The resulting perfusion images preserve high spatial fidelity while capturing rapid perfusion changes, facilitating investigation of functional dynamics and the implementation of complex fMRI paradigms.

Recent improvement in UHF MRI hardware, achieved by the NexGen 7 T scanner with its unprecedented gradient performance and high density transmit and receiver coil arrays, have significantly pushed the boundaries of spatiotemporal resolution in neuroimaging [42]. These developments have enabled whole-brain BOLD and VASO imaging at submillimeter resolution, facilitating the study of laminar-specific neural dynamics previously inaccessible at conventional resolutions. Our study spearheads the ASL sequence on the NexGen 7 T scanner by deliberately designed pulse sequence to capitalize on the hardware performance: (1) the spiral gradient waveforms maintain high gradient amplitudes following a rapid ramp-up; (2) highly incoherent undersampling strategies leverage the high density receiver array to minimize g-factor related noise amplification; (3) higher spatial resolution supported by higher signal at 7 T rather than 3 T. Compared to BOLD and VASO, high-resolution ASL offers unique advantages, including quantitative measurements of CBF, high spatial specificity without draining vein effect, and a direct physiological link to neuronal activities.

The past two decades have seen dramatic progress in understanding the complex hierarchical organization of the human brain using a range of imaging technologies. It is now possible to reconstruct detailed synaptic connections between neurons in brain tissue samples (~$1mm^3$) using microscopy and AI based analyses [60, 61]. Meanwhile, large field-of-view imaging techniques such as BOLD fMRI and diffusion MRI with improved spatiotemporal resolutions have been applied widely to study the connectome and dynamics of macroscopic brain networks, as demonstrated by the Human Connectome Project (HCP) [62]. Despite these parallel advances in neuroimaging, it remains challenging to bridge mesoscopic MRI with the underlying cellular structure and function that are typically achieved through invasive imaging and/or post-mortem studies. In this study, the strong correlation between in vivo laminar CBF profiles and ex vivo CSI suggests that perfusion closely reflects underlying cell body densities. The potential of laminar CBF as an in vivo surrogate for cellular structure make it possible to (1) detect early cortical atrophy or selective laminar loss (e.g., in Alzheimer's or frontotemporal dementia) before gross anatomical changes appear [63]; (2) combine with mitochondrial density or metabolism maps (e.g., MitoBrainMap [64]) to evaluate layer-specific neuroenergetics.

The proposed technology offers a noninvasive imaging approach for understanding the mechanism underlying "task-negative" activity in the DMN and "task-positive" activity in frontoparietal networks during cognitively demanding tasks, such as WM, which was currently only possible with invasive intracranial electrophysiological recording in limited brain regions [65]. Detecting task-negative BOLD responses is inherently challenging due to the complex interplay between CBF, CBV, and $CMRO_2$. Unlike task-positive activation, where CBF increases typically exceed the rise in $CMRO_2$ and produce strong BOLD signals, deactivations involve concurrent decreases in CBF and $CMRO_2$ that may not align proportional, contributing to reduced sensitivity and interpretability of BOLD deactivation [66]. In contrast, ASL directly measures CBF and is thus more sensitive to both increases and decreases in perfusion. In the WM experiments, we consistently observed higher relative ΔCBF in the deep layer than in the superficial layer across both task-positive and task-negative regions, replicated in all five subjects and on different scanner systems. This reproducibility demonstrates the method's capability to reliably capture distinct laminar neural activity, although the biological interpretation of this deep-layer predominance warrants further investigation in future studies.

Several limitations should be acknowledged in this study. First, the sample size for each experiment was small and demographically homogeneous. Second, although a standard PLD was employed, arterial transit time (ATT) was not directly measured. While the inclusion of young healthy subjects minimizes potential ATT effect, we acknowledge that ATT and T1 differences across cortical layers may introduce subtle bias in laminar CBF. The current sequence framework could be extended to incorporate multiple PLDs dynamic frames [67], enabling simultaneous mapping of ATT and CBF in future studies to further refine laminar perfusion quantification. Third, while MFI was applied to mitigate off-resonance blurring artifacts, small residual errors due to off-resonance effects may persist. At 7 T, concomitant field is negligible, but the impact of B_0 drift may become significant, especially in fMRI studies involving multiple runs. Fourth, the reconstruction time was substantially extended by MFI. Emerging methods such as Toeplitz-based non-Cartesian reconstruction may help accelerate processing [68, 69]. Deep learning approaches hold promise for addressing blurring artifacts arising from both off-resonance and ASL signal decay [70–72]. Additionally, LE may be further improved by dynamic B0 shimming and parallel-transmission-enabled B1 shimming, and SAR of each sequence component can be reduced [73, 74]. Physiological noise may be further reduced by applying a stronger BS; however, at 7 T, this remains constrained by SAR limits and cumulative ASL signal loss due to imperfect inversions, necessitating careful trade-offs between tissue suppression and perfusion signal preservation. Moreover, our comparisons did not evaluate alternative reconstruction sparsity bases (e.g., wavelets) or alternative Cartesian view-ordering schemes such as CASPR [75], which may yield different trade-offs in image sharpness and motion robustness. Finally, the clinical significance was not assessed in patient populations, which merits future evaluation.

5 | Conclusion

The next generation ASL presented in this study achieved robust mesoscale whole-brain perfusion imaging at 7 T MRI. Our method offers a noninvasive approach to bridge mesoscopic MRI with microscopic cellular imaging, and to investigate neural excitation and inhibition underlying fMRI activations.

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RF1AG084072, S10-OD025312, U01-EB025162, U24-NS129949, UF1-NS100614).

Conflicts of Interest

Samantha J. Ma is an employee of Siemens Healthcare.

Data Availability Statement

Experimental data are available upon request at http://www.loft-lab.org/index-5.html. The pulse sequence and reconstruction algorithm described in this work can be requested from the corresponding author through Siemens C2P (Customer to Peer) and University Southern California (USC) Material Transfer Agreement (MTA).

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